Abstract:

Groundwater is a critical component of water security, particularly in rapidly urbanizing regions where increasing population, land-use changes, irregular rainfall, and intensive groundwater extraction place continuous pressure on subsurface water resources. Conventional groundwater assessment methods often depend on field measurements and spatial analysis of individual environmental factors, making continuous, data-driven assessment difficult over large and heterogeneous areas. This project proposes an AI-based groundwater potential prediction framework for Bengaluru, Karnataka, integrating remotely sensed, terrain, rainfall, and groundwater-level information into a unified machine-learning workflow.

The proposed system combines Sentinel-2 multispectral satellite imagery, terrain information derived from a Digital Elevation Model (DEM), historical rainfall observations, and observed groundwater-level data to construct a spatially and temporally aligned feature dataset. Sentinel-2 imagery is used to derive surface characteristics and spectral indicators such as Normalized Difference Vegetation Index (NDVI) and Normalized Difference Water Index (NDWI), while the DEM provides terrain-related variables such as elevation and slope. Rainfall data obtained from satellite-based observations are incorporated to represent the temporal influence of precipitation on groundwater conditions. Groundwater observations from the available Karnataka groundwater monitoring dataset covering 2021–2025 provide the reference target for model development and validation.

The integrated dataset is processed through systematic geospatial preprocessing, quality control, feature extraction, temporal alignment, and normalization/standardization where required before being supplied to a Light Gradient Boosting Machine (LightGBM) model. The model is intended to learn the relationship between environmental and remotely sensed variables and observed groundwater conditions, enabling spatial prediction across the Bengaluru study region. The resulting predictions can be transformed into a groundwater potential/prediction map, providing a spatial representation that can support identification of areas exhibiting different groundwater conditions and assist subsequent water-resource assessment.

The proposed framework emphasizes the integration of Earth observation, environmental data, geospatial feature engineering, and artificial intelligence

rather than relying on a single groundwater indicator. Its modular design also permits the future incorporation of additional factors such as land-use/land-cover, drainage characteristics, soil, geological information, and additional satellite observations. The project therefore establishes a data-driven foundation for groundwater assessment in Bengaluru and provides a reproducible framework that can be extended to other regions where suitable groundwater observations and environmental datasets are available.